

Demand Analysis

Module 2

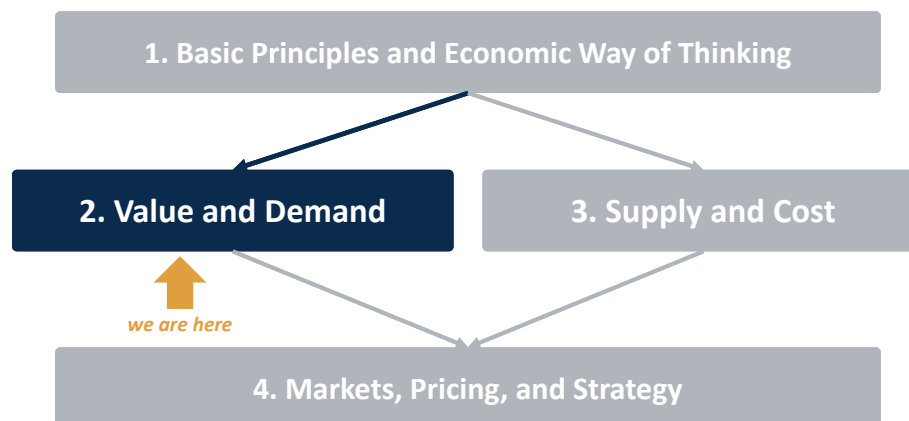
Management 405

Prof. Nico Voigtländer · UCLA Anderson

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Module 2 · Course Roadmap

Agenda for the Class



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Module 2 · Agenda

Outline of Module 2

- | | | |
|-----------|--|-------------------|
| 1 | The Law of Demand
When the price falls, quantity demanded rises – holding everything else constant | In class |
| 2 | Elasticities
How strongly demand responds to price, income, and the prices of other goods | In class |
| 3 | Demand and Revenue
How the demand curve translates into total revenue | Videos 1+2 |
| 3a | Elasticity and Revenue
When a price increase raises revenue – and when it backfires | Video 1 |
| 3b | Marginal Revenue
The extra revenue from selling one more unit | Video 2 |
| 4 | Demand Estimation
Measuring demand from data – market experiments and regression | Video 3 |

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Module 2 · Agenda

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The Law of Demand

Crucial assumption:

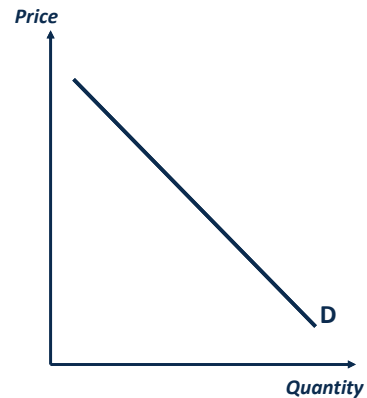
- Holding everything else constant (income, weather, product attributes, perception of product quality ...)
- In econ parlance: “ceteris paribus” (**c.p.**)

The Law of Demand says:

- As the price of a good declines, quantity demanded increases
 - i.e., the demand curve is downward-sloping

Reasons: As the price declines...

- More customers are willing to buy the product
- Existing customers buy more



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1. MORE Customers Buy the Product

- How much are you willing to pay for a slice of (great) pizza when you are hungry?
- Maximum willingness to pay is your **Marginal Personal Value (MPV)** of the slice



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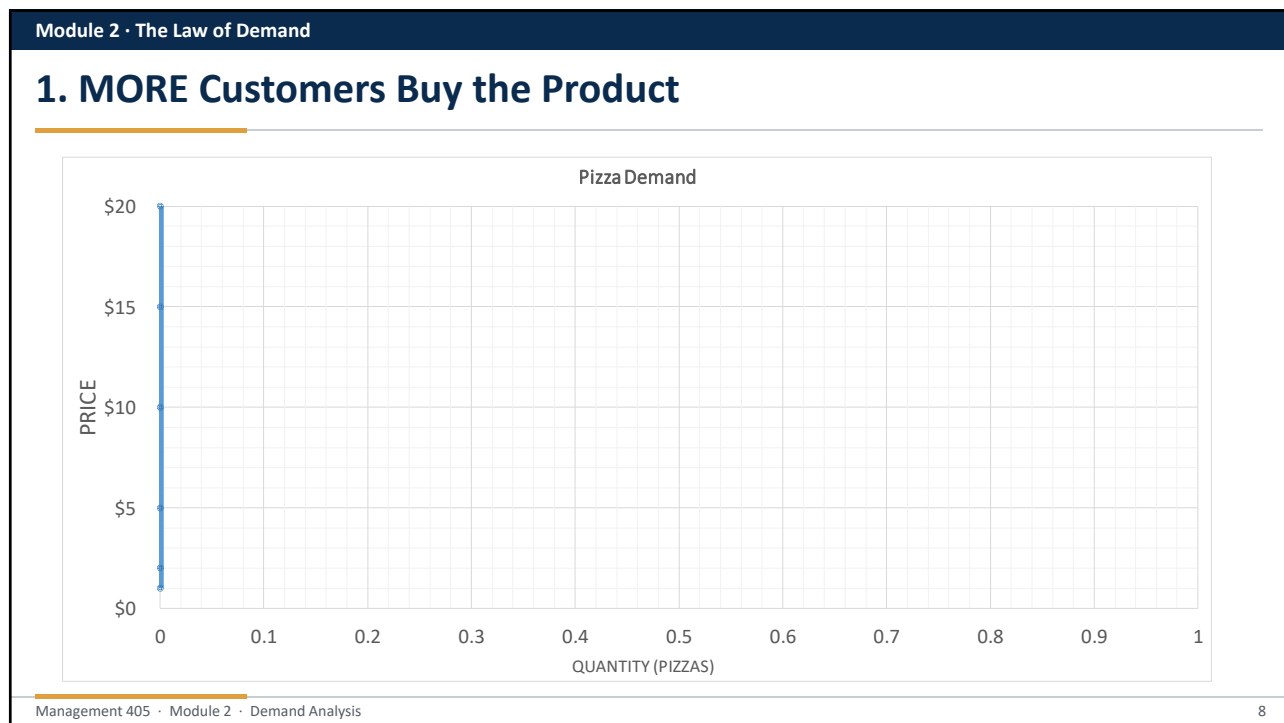
How much are you willing to pay for a slice of (great) pizza when you're hungry?

0

\$20	0
\$15	0
\$10	0
\$5	0
\$2	0
\$1	0
zero	0

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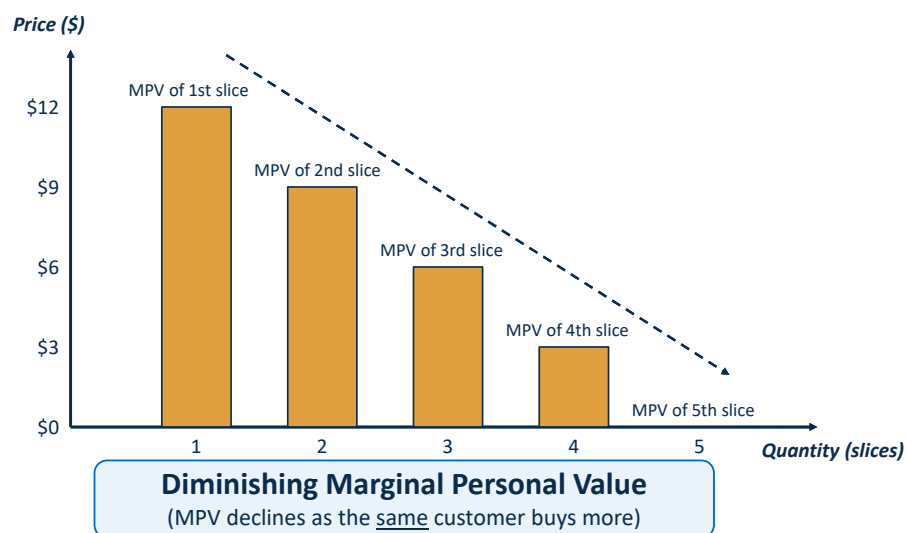
2. Existing Customers Buy MORE

- What about a second slice of pizza?
 - A third?
 - A fourth?
 - A millionth?
- ... Diminishing Marginal Personal Value (MPV)!



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To the Same Consumer, Each Extra Slice Is Worth Less



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Diminishing Marginal Personal Value

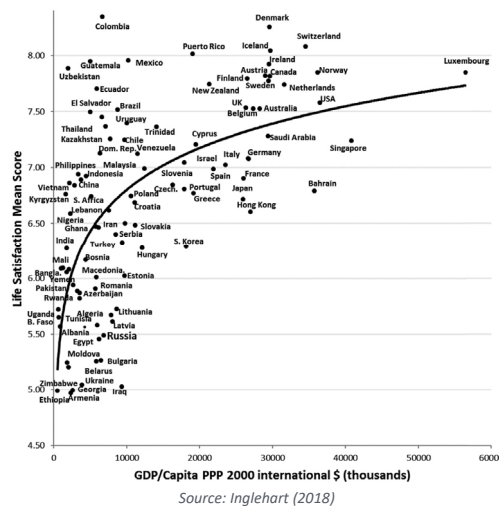
“ I can understand wanting to have a million dollars... but once you get beyond that, I have to tell you, it's the same hamburger.

– Bill Gates



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Diminishing MPV: Income and Life Satisfaction



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Consumer Optimization

General optimization rule (from Module 1)

$$MB = MC$$

MB: marginal benefit of an extra unit of a good/service

MC: marginal cost, which includes

- the price of the good or service
- the opportunity cost (e.g. time spent getting a haircut)

In the context of consumption we use MPV
(essentially the same as MB)

Choose consumption quantity where:

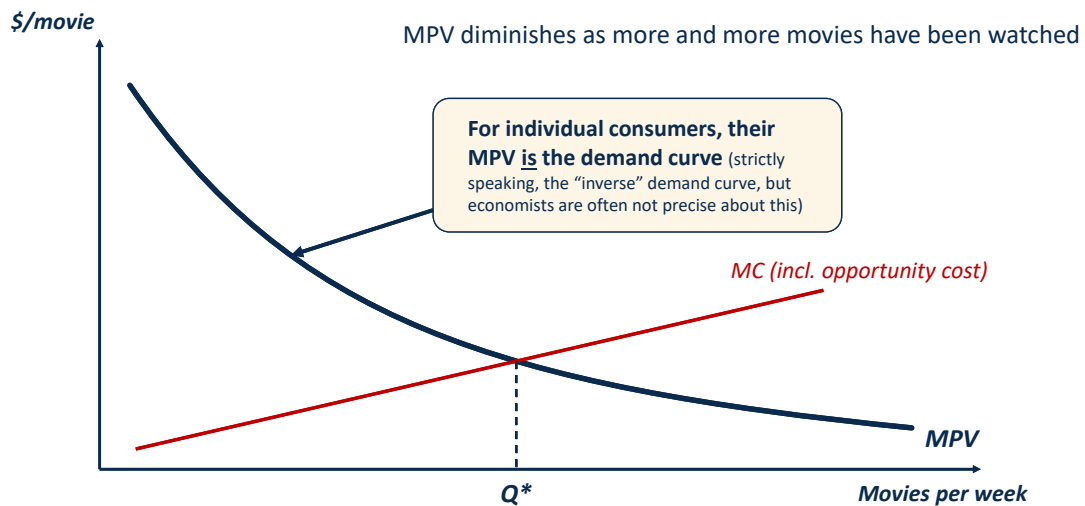
$$MPV = MC$$

If $MPV > MC$: buy more

If $MPV < MC$: buy less

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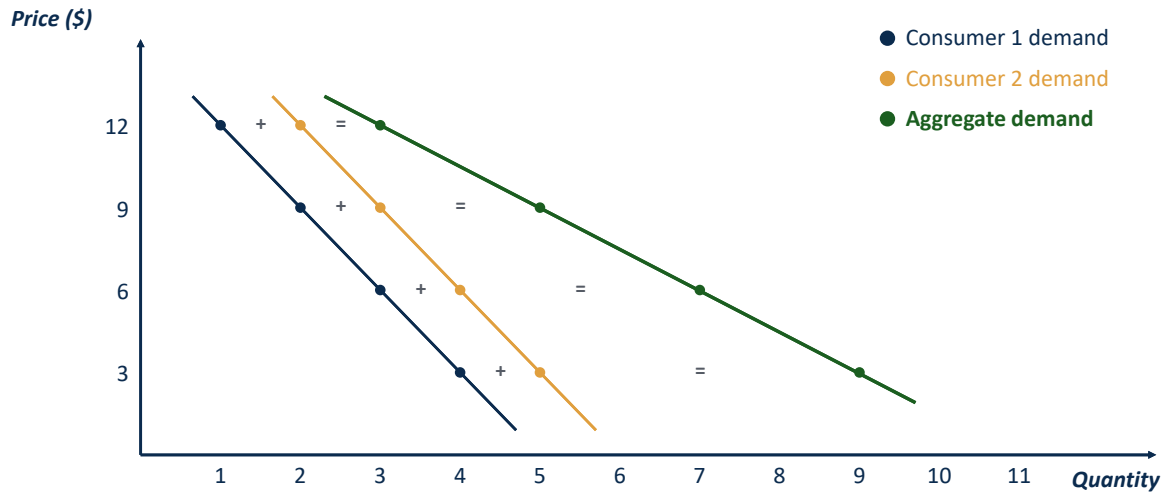
Optimal Number of Movies to Watch (per Week, by One Customer)



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Module 2 · The Law of Demand

Aggregating Individual Consumers' Demand



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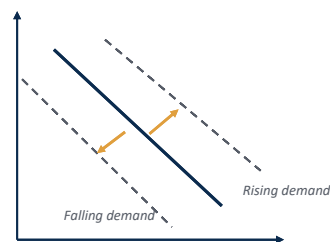
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Module 2 · The Law of Demand

Factors Affecting Demand

- Income
- Prices of related goods
 - Complements versus substitutes
 - Complements can be used strategically
 - E.g., Tesla & Superchargers. Ryanair.
- Price expectations
- Population size and composition
- Product quality, advertising, placement (*within the firm's control*)
- Network effects



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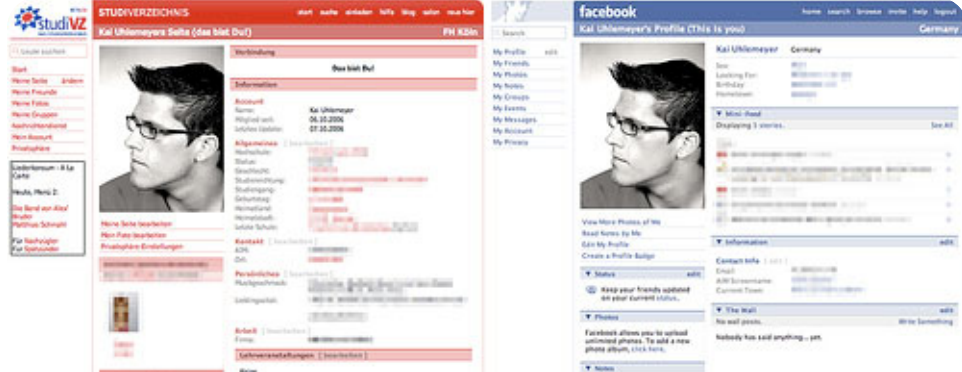
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Module 2 · The Law of Demand

Implications of Positive Network Effects

- Network effects make switching more difficult for customers
- “Tippy” Markets
- Winner-take-all



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Module 2 · The Law of Demand

The Snob Effect (Negative Network Effects)

Snob effect: exclusivity is part of the value, so demand falls as more people own the good

The Wild Economics Behind Ferrari's Domination of the Luxury Car Market

Limiting production is helping to make its sports cars coveted—and the company the most valuable automaker in Europe



Trying to Dethrone the Birkin? Make Fewer Bags

Chanel has taken a less effective approach to competing with the Hermès bag than American luxury brand The Row



Source: The Wall Street Journal (2025)

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Module 2 · The Law of Demand

Remember About Demand

- Represents the willingness to pay by all actual or potential customers for a good/service
- Price changes while everything else is kept constant (c.p.)
- We usually represent demand as a line, for simplicity
- We draw P on the vertical axis, but mathematically:
 - $Q = 10 - P$: this is the demand function (Q as function of P)
 - $P = 10 - Q$: is the “inverse” demand function → so strictly speaking, we draw the “inverse” demand function



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Module 2 · Agenda

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Module 2 · Elasticities

Generic Definition of Elasticity

- Refers to the **responsiveness to external forces**:
 - Elastic**: changes when punched
 - Inelastic**: stays in original position when punched
- What's the “punch” in economics?

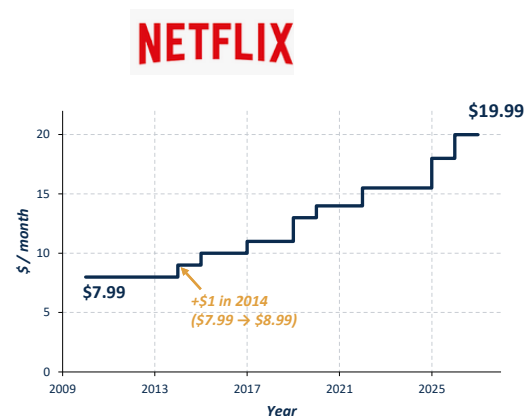


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We Use Elasticity to Study Effects of Price Increases

- Netflix frequently raises its monthly subscription price
- What are the implications of price increases for:
 - Number of subscriptions?
 - Revenues?
- Key concept: Elasticity



Netflix U.S. Standard plan · Source: Netflix price announcements

► **Module 2, Practice Video 1**
numerical example with Netflix price increase

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What Is the Elasticity of Demand?

$$E = \frac{\% \Delta Q}{\% \Delta X}$$

percentage change

- **Definition:** How quantity demanded (Q) changes with respect to another variable (X)
– E.g., X = price, income...
- **Characteristic:** **Unit free**
- **How to talk about elasticity:** A % change in quantity demanded (Q) due to a 1% change in another variable (X)

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Three Types of Elasticity

Own-price elasticity

$$E_D = \frac{\% \Delta Q}{\% \Delta P}$$

Income elasticity

$$E_I = \frac{\% \Delta Q}{\% \Delta I}$$

Cross-price elasticity

$$E_{X,Y} = \frac{\% \Delta Q_X}{\% \Delta P_Y}$$

All three follow the same recipe: a % change in quantity, divided by the % change in the driving variable

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Own-Price Elasticity

$$E_D = \frac{\% \Delta Q}{\% \Delta P}$$

- % change in quantity demanded, divided by the % change in price (of the same product)
- **Intuition:** How sensitive demand is to changes in the own price of the good
- Sign of E_D : **Negative** (Law of Demand: $P \uparrow \Rightarrow Q \downarrow$)

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Example: Demand Elasticity for Water



Consulting company estimated: price elasticity of demand for water in L.A. is -0.4



LADWP wants to reduce quantity demanded by 10%



By how much should it raise its price?

Poll Break

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By how much should LADWP raise the price of water?

(A) +10%	0%
(B) +0.4	0%
(C) +25%	0%
(D) +15%	0%
(E) None of the above	0%

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Categories of Own-Price Elasticity

$$E_D = \frac{\% \Delta Q}{\% \Delta P}$$

absolute value

$0 < E_D < 1$ $-1 < E_D < 0$ demand is inelastic	$ E_D > 1$ $E_D < -1$ demand is elastic	$ E_D = 1$ $E_D = -1$ demand is unit elastic
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Example: Demand Elasticity for Yoga Lessons

- CorePower Yoga cuts its price by 17%
- The number of booked lessons **double** as a result
- What is the price elasticity of demand that the yoga studio faces?



Poll Break

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What is the implied price elasticity of demand that the yoga studio faces?

- | | |
|----------------|----|
| (A) About -6 | 0% |
| (B) About -0.6 | 0% |
| (C) About -1.7 | 0% |
| (D) About -17 | 0% |
| (E) About 6% | 0% |

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Computing Elasticity: **Method 1**

- When you have two price/quantity observations: (P_0, Q_0) and (P_1, Q_1)

$$E_D = \frac{\% \Delta Q}{\% \Delta P} = \frac{\frac{Q_1 - Q_0}{Q_0}}{\frac{P_1 - P_0}{P_0}}$$

CONVENTION
always relative to the initial
point (P_0, Q_0)

This method **approximates** the percentage changes

► TA Math Review Videos

✎ Problem Set 2

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Lottery Sales: Mega Millions Raises Its Price

- April 2025: Mega Millions raises its ticket price from \$2 to \$5
- As a result, in Massachusetts, weekly ticket sales fell from about 142,000 to about 100,000
- What is the implied price elasticity of demand** (using \$2 as the initial price)?

Source: Massachusetts Lottery, via WBZ-TV / WROR, April 2025

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Poll Break

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What is the Demand Elasticity for Mega Millions Lottery Tickets?

- About 2 0%
- About -5 0%
- About -0.2 0%
- About 0.2 0%
- About -1 0%

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Computing Elasticity: **Method 2** ("Point Elasticity")

- When you know the entire demand curve, use it!

$$E_D = \frac{\% \Delta Q}{\% \Delta P} = \frac{\frac{\Delta Q}{Q}}{\frac{\Delta P}{P}} = \left(\frac{\Delta Q}{\Delta P} \right) \cdot \frac{P}{Q}$$

*Slope of the demand curve
(constant for a linear demand curve)*

- $\Delta Q / \Delta P$ is the slope of the demand curve
 - Mathematically: Derivative of Q with respect to P : dQ/dP
 - See TA Math Review videos

This method computes the **exact** elasticity at a given (P, Q) point

 **Problem Set 2**

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Computing the Point Elasticity: **Method 2** – Step-by-Step

Demand function: $Q = 400 - 4P$

Task: Compute the point elasticity at $P = 25$

Step 1: Compute derivative dQ/dP

$$\frac{dQ}{dP} = -4$$

Step 2: Obtain Q at $P = 25$

$$Q = 400 - 4 \times 25 = 300$$

Step 3: Put everything into the formula

$$E_D = \frac{dQ}{dP} \cdot \frac{P}{Q} = -4 \times \frac{25}{300} = -\frac{1}{3}$$

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Obtain Price Elasticity from the Demand Function

A firm's demand function is $Q = 10 - P$

The firm charges a price of $P = 2$

What is its elasticity of demand at this point?

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If $Q=10-P$, what is the elasticity at $P=2$?

0

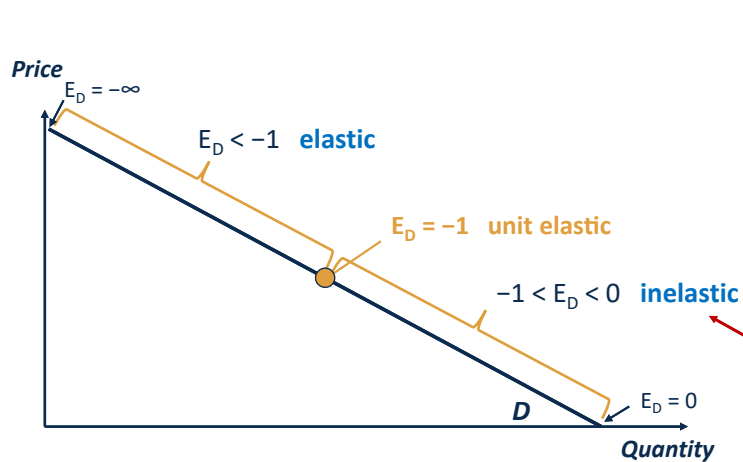
- (A) -2 0%
- (B) -0.25 0%
- (C) -0.67 0%
- (D) -4 0%
- (E) -0.25% 0%

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Elasticity Along the Demand Curve: Linear Case



$$E_D = \frac{\% \Delta Q}{\% \Delta P} = \frac{\Delta Q}{\Delta P} \cdot \frac{P}{Q}$$

Linear demand curve:

E_D goes from $-\infty$ (approaching y-axis) to 0 (approaching x-axis)

We will learn: **Firms should NOT operate in the inelastic area** (Module 2, Videos 1+2)

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Important Insight

Firms that operate in the inelastic part of demand **should raise their price**

- **Intuition:** With inelastic demand, quantity reacts little when prices are raised → higher prices will raise revenues
 - Also, quantity declines, so (variable) costs decline
 - Thus: Firms can increase profits by raising prices
- **Example:** A study in 2016 estimated Uber's price elasticity as -0.4 (inelastic)
 - Subsequently, Uber raised its prices by an average of 18% per year in the 2018-2022 period. Much more than average inflation over the same period (4.5%)
 - Now prices (and revenues) are much higher

► Elaborated in
Module 2, Videos 1+2



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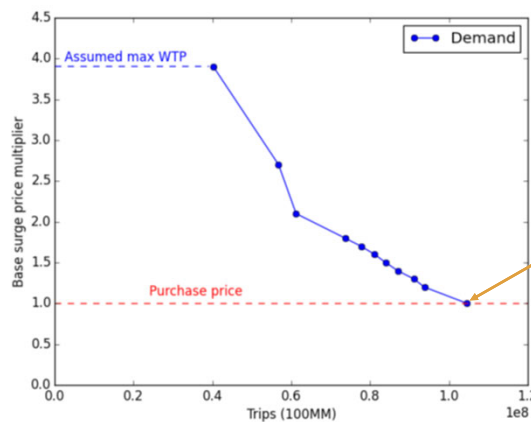
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Estimated Demand Curve for Uber (Surge Prices, 2016)

If there's demand, I can provide detail on this estimation in a 'Coffee & Econ' session – but this is not relevant for the exam



Elasticity = -0.4 at 2016 base price



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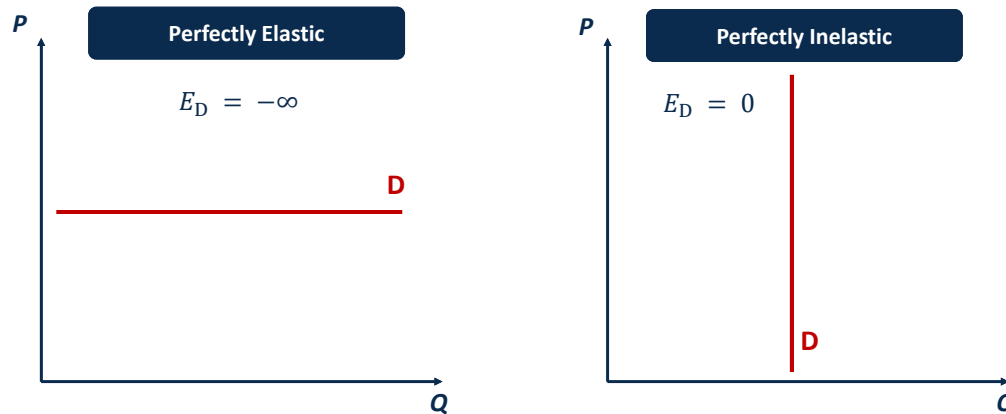
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Special Cases of Price Elasticities

- Elasticity changes along the demand curve, except ...



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What Determines the Price Elasticity of Demand for a Product?

- Availability of substitutes
 - Number
 - Closeness
- Willingness to switch



Product	Estimated Own-Price Elasticity of Demand
BROAD FOOD GROUPS	
Eggs	-0.06
Beef	-0.35
Fish	-0.39
Juice	-1.05
SPECIFIC BREAKFAST CEREALS	
Cap'n Crunch	-2.28
Froot Loops	-2.34
Kellogg's Corn Flakes	-3.38
Cheerios	-3.66
Shredded Wheat	-4.25
SPECIFIC AUTOMOBILES	
Jeep Grand Cherokee	-3.06
Cadillac Seville	-3.16
Toyota Corolla	-3.92

Source: food data – Huang & Lin (2000), USDA Technical Bulletin 1887

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Market vs. Firm Elasticity

- The elasticity a company faces is **not the same** as the elasticity the industry faces
- A firm's demand is more elastic when...
 - the overall market elasticity is high
 - the firm's market share is small
 - competitors don't react to its price changes

4 Gas stations at Slauson Ave and La Brea Ave, Los Angeles



Demand at individual gas stations is much more elastic than the overall market demand for gas

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Module 2 · Elasticities

Cheat Sheet: Own-Price Elasticity

Definition

$$E_D = \frac{\% \Delta Q}{\% \Delta P}$$

Intuitively: measures consumers' price sensitivity

Categories

Inelastic: $-1 < E_D < 0$ ($|E_D| < 1$)

Unit-elastic: $E_D = -1$

Elastic: $E_D < -1$ ($|E_D| > 1$)

Computing elasticity

Two observed points (approximation):

$$E_d = \frac{(Q_1 - Q_0)/Q_0}{(P_1 - P_0)/P_0}$$

Whole demand function known:

$$E_d = \frac{\Delta Q}{\Delta P} \cdot \frac{P}{Q}$$

$\Delta Q / \Delta P = \text{slope of the demand function (Q as a function of P)}$

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Next: Other Elasticities

Own-price elasticity

$$E_D = \frac{\% \Delta Q}{\% \Delta P}$$

Income elasticity

$$E_I = \frac{\% \Delta Q}{\% \Delta I}$$

Cross-price elasticity

$$E_{X,Y} = \frac{\% \Delta Q_X}{\% \Delta P_Y}$$

Own-price elasticity ✓ — next: how demand responds to income, and to other goods' prices

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Income Elasticity

$$E_I = \frac{\% \Delta Q}{\% \Delta I}$$

- % change in quantity demanded, divided by the % change in **income (I)**
- **Intuition:** How sensitive demand is to changes in customer's income
- If $E_I > 0 \rightarrow$ **normal good** (if $E_I > 1 \rightarrow$ **luxury good**)
- If $E_I < 0 \rightarrow$ **inferior good**



normal



luxury



inferior

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Example: Income Elasticity

- Average income in the U.S. goes up by 2%
- As a result, U.S. demand for the Rivian R3 goes up by 5%
- **What is the implied income elasticity of demand for the Rivian R3?**

**Poll Break**

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What is the implied income elasticity of demand for the R3?

0

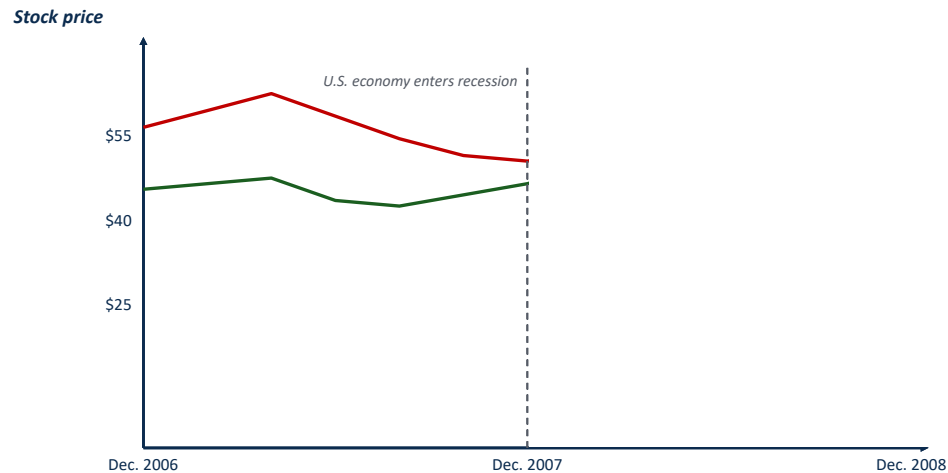
- | | | |
|----------|-------------|----|
| (A) 0.4 | <div></div> | 0% |
| (B) 1 | <div></div> | 0% |
| (C) 2.5 | <div></div> | 0% |
| (D) -2.5 | <div></div> | 0% |
| (E) 4 | <div></div> | 0% |

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Which Retailers Do Well in a Recession? Clues from Income Elasticity



Source: Stevenson/Wolfers, *Principles of Economics*, 1e, © 2020 Worth Publishers (Figure 9); series approximate

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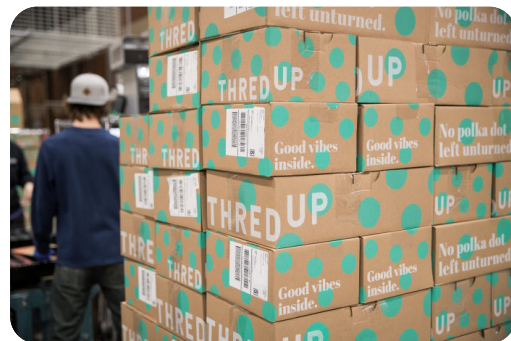
Inferior Goods in the News

THE WALL STREET JOURNAL

ThredUp Gains Shoppers Weary of Economic Uncertainty

The number of active buyers rose 17%, to 1.47 million, in the second quarter.

Inferior goods and recessions: Demand rises when incomes fall. Secondhand clothing, discount grocers, store brands and public transit all grow in a downturn



Source: The Wall Street Journal (2025)

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Cross-Price Elasticity

$$E_{X,Y} = \frac{\% \Delta Q_X}{\% \Delta P_Y}$$

- % change in the quantity demanded of good X, divided by the % change in the price of another good Y
- **Intuition:** How sensitive demand is to changes in the price of another good
- Two cases:
 - If $E_{X,Y} > 0$, then the goods are **substitutes**
 - If $E_{X,Y} < 0$, then the goods are **complements**



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Example: Movie Ticket and Popcorn

- When a movie theater *increased* its ticket price from \$15 to \$18, the quantity demanded of popcorn *decreased* by 8%
- **What is the implied cross-price elasticity?**



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Poll Break

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The cross-price elasticity of popcorn with respect to movie tickets is...

(A) 5% 0%

(B) 0.2 0%

(C) 0.4 0%

(D) -2 0%

(E) -0.4 0%

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Cross-Price Elasticity in the News

THE WALL STREET JOURNAL.

Sticker Shock at the Pump Fuels a Surge in Hybrid Sales

Sales of hybrid cars carried the day in May as buyers seek better fuel economy

As the price of fuel goes up, demand for fuel-powered vehicles goes down
→ The two are **complements**

Demand then shifts to a **substitute** for fuel-powered vehicles: electric vehicles, as electricity has become **relatively cheaper**




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Module 2 · Elasticities

Estimated Cross-Price Elasticities: Ready-to-Eat Cereal






	Corn Flakes	Frosted Flakes	Rice Krispies
Frosted Flakes	0.15		
Rice Krispies	0.19	0.14	
Froot Loops	0.02	0.13	0.04

Larger cross-price elasticity → closer substitutes

Source: Nevo, A. (2001). Measuring Market Power in the Ready-to-Eat Cereal Industry. *Econometrica*, 69(2), 307–342

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Module 2 · Agenda

Outline of Module 2: Upcoming Videos 1+2

- 1 The Law of Demand In class
- 2 Elasticities In class
- 3 Demand and Revenue Videos 1+2
 - How the demand curve translates into total revenue
 - 3a Elasticity and Revenue Video 1
 - When a price increase raises revenue – and when it backfires
 - 3b Marginal Revenue Video 2
 - The extra revenue from selling one more unit
- 4 Demand Estimation Video 3

► Module 2 Videos 1+2 · Practice Videos 1+2
On BL under “Module 2 Post-Work”

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Outline of Module 2: Upcoming Video 3

1 The Law of Demand

In class

2 Elasticities

In class

3 Demand and Revenue

Videos 1+2

3a Elasticity and Revenue

Video 1

3b Marginal Revenue

Note: You do not need to perform the actual estimation (regression). But you need to understand how to interpret regression coefficients

▶ Module 2 Video 3 · 📄 Problem Set 2
On BL under “Module 2 Post-Work”

4 Demand Estimation

Video 3

Measuring demand from data – market experiments and regression